

4.4.1 Consider the following Turing machine

$M =$ “on input $\langle A \rangle$, where A is a DFA

1. Create a DFA B such that $L(B) = \Sigma^*$
2. Submit $\langle A, B \rangle$ to the decider for EQ_{DFA}
3. If it accepts, **accept**
4. If it reject, **reject**.”

M is clearly a decider since steps 1, 3, and 4 will not create an infinite loop and step 2 calls a decider. Furthermore, M will accept those DFA's whose language is the same as B 's language – i.e. DFA's whose language is Σ^* -- and will reject all other languages. Therefore, M is a decider for ALL_{DFA} so ALL_{DFA} is decidable.

4.12 Consider the following Turing machine

$M =$ “On input $\langle R, S \rangle$ where R and S are regular expressions

1. Convert R to DFA A and S to DFA B
2. Construct DFA C such that $L(C) = L(B) \cap L(A)$
3. Submit $\langle A, C \rangle$ to the decider for EQ_{DFA}
4. If it accepts, **accept**
5. If it rejects, **reject**.”

M is a decider since steps 1, 2, 4, and 5 will not create and infinite loops and step 2 calls a decider. Also, M accepts $\langle R, S \rangle$ iff $L(R) = L(R) \cap L(S)$ – i.e., iff $L(R) \subseteq L(S)$. Therefore, M is a decider for A so A is decidable.

4.19.1 Note that if the DFA accepts w^R whenever it accepts w , then $L(M) = L(M^R)$, where M^R is the DFA that accepts the reverse of strings accepted by M . In a previous homework, we proved that if M is a regular language, then so is M^R .

Consider the following Turing machine

$T =$ “On input $\langle M \rangle$, where M is a DFA

1. Construct DFA N that accept the reverse of strings accepted by M
2. Submit $\langle M, N \rangle$ to the decider for EQ_{DFA}
3. If it accepts, **accept**
4. If it rejects, **reject**.”

T is a decider since 1, 3, and 4 will not create an infinite loop and step 2 calls a decider. Also, T accept M iff $L(M) = L(M^R)$. Therefore T decides S so S is decidable.